SC26 Figures 3–6: reproduction outcome

Environment: clariden-ln004 · fresh run output: /iopsstor/scratch/cscs/btommaso/sc26_reproduction_20260722/reproduction_clariden-ln004_20260722T013845

Figure	Paper panel	Input evidence	Fresh result in this run
3	AllReduce 128 MiB, 16N/64 GPU, ch1 + auto	exact original NSYS	raw NSYS → fresh SQLite → fresh GOAL/metadata → LGS/LP
4	Mixed20, 16N/64 GPU, ch1	two raw candidates	old job reproduces curve; corrected written-config campaign does not
5	Llama N4/GPU16/DP16	related static BIN + old metadata	paper curve not reproduced; raw trace/GOAL absent; 7B/8B conflict
6 Llama	N32/GPU128/DP128	derived metadata only	fresh Composite latency/bandwidth; paper says 70B, metadata says 7B
6 vLLM	Llama-3.1-8B, N2/GPU8, 128 tokens	original expired	not rerunnable; displayed as blocked, never substituted with 70B

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Match against packaged paper/reference CSVs:
• Fig. 3 ch1: latency Monolithic 0.078% max / 0.025% mean; bandwidth Monolithic 2.373% max / 0.230% mean
• Fig. 3 auto: latency Monolithic 0.166% max / 0.056% mean; bandwidth Monolithic 1.493% max / 0.523% mean
• Fig. 3 LGS: ch1 latency 5.256% max / 0.646% mean; auto latency 2.285% max / 0.548% mean
• Fig. 4 old curve-source: Monolithic 0.118% max / 0.060% mean; LGS 2.076% max / 0.612% mean
• Fig. 4 corrected raw campaign: Composite 297.392% max / 196.042% mean; Monolithic 306.871% max / 248.451% mean
• Fig. 5 old metadata Composite vs packaged paper curve: 15.910% max / 10.038% mean
• Fig. 6 Llama metadata: latency 0.128% max / 0.057% mean; bandwidth 0.358% max / 0.026% mean
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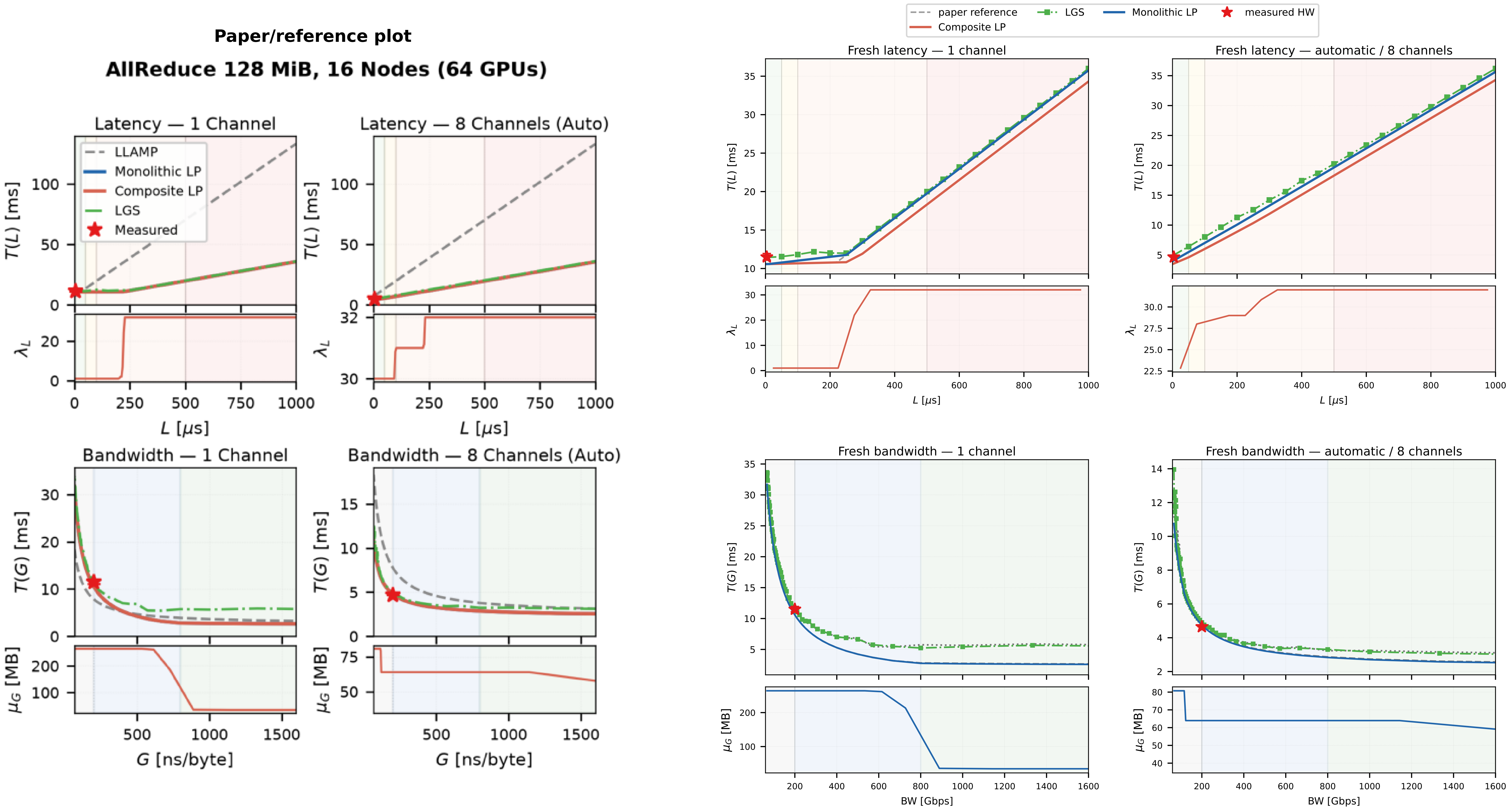
Left-hand plots on later pages are regenerated from the artifact's packaged paper data. They are references, not independent reruns. Right-hand plots use only outputs generated during this run. Gray dashed curves on the right are paper CSVs shown solely for visual comparison.

Figure 3 — 128 MiB AllReduce

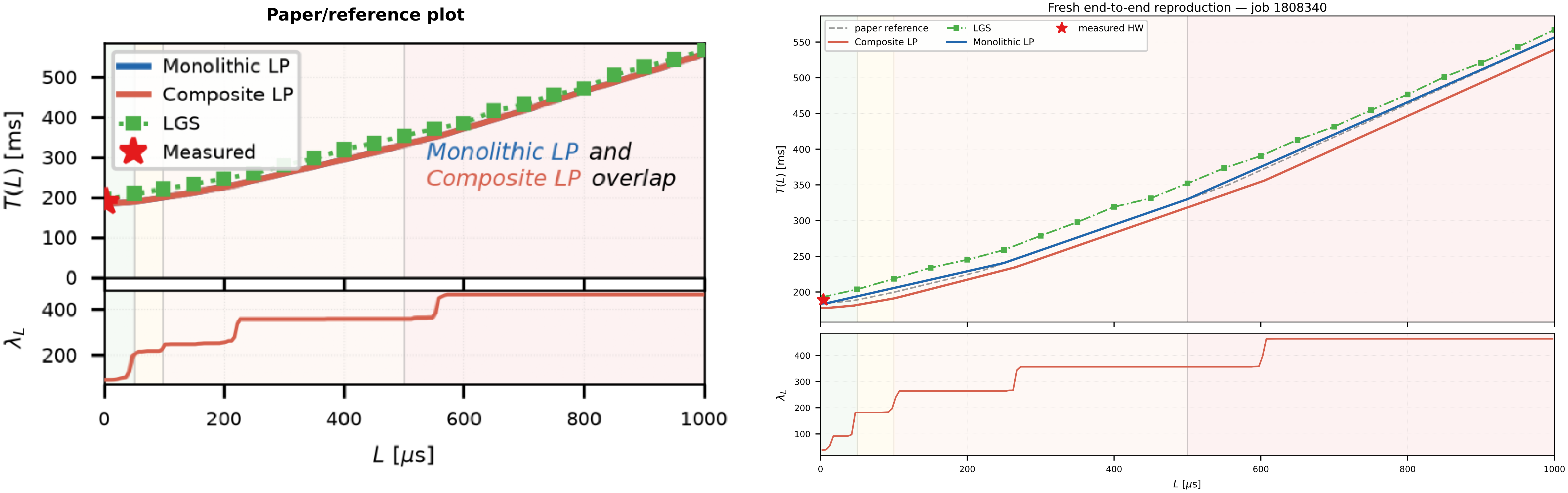
Paper/reference on the left · fresh raw-input reproduction on the right



Evidence: 64 original NSYS reports per mode; 16 nodes, 64 ranks, Ring/Simple. The recovered historical generator establishes four physical NIC queues per GH200 node. Monolithic bandwidth: ch1 0.230% mean error; auto 0.523% mean error. All right-hand solid/marker curves are fresh.

Figure 4 — mixed collectives

Paper/reference on the left · fresh end-to-end NSYS reproduction on the right



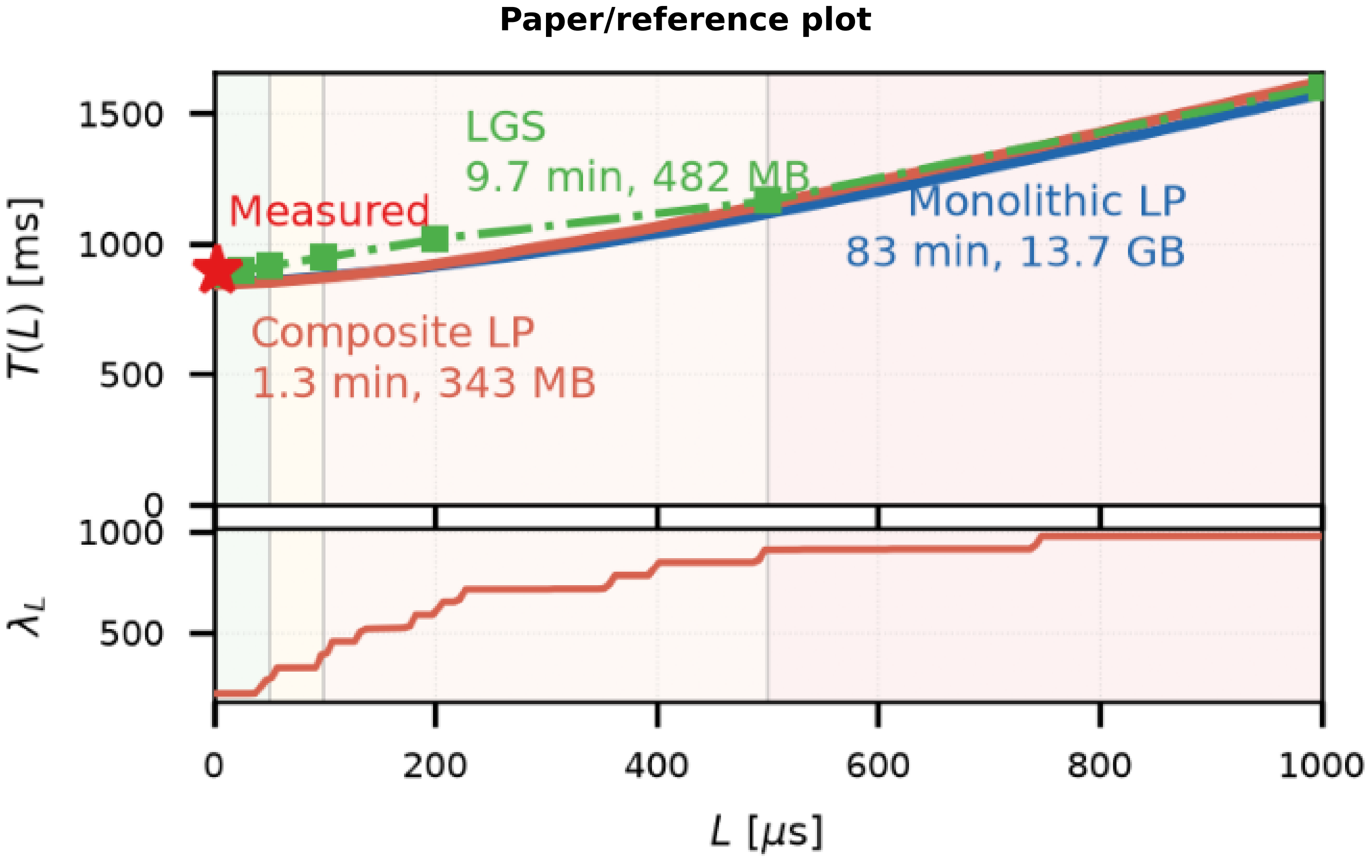
Provenance check (kept separate)

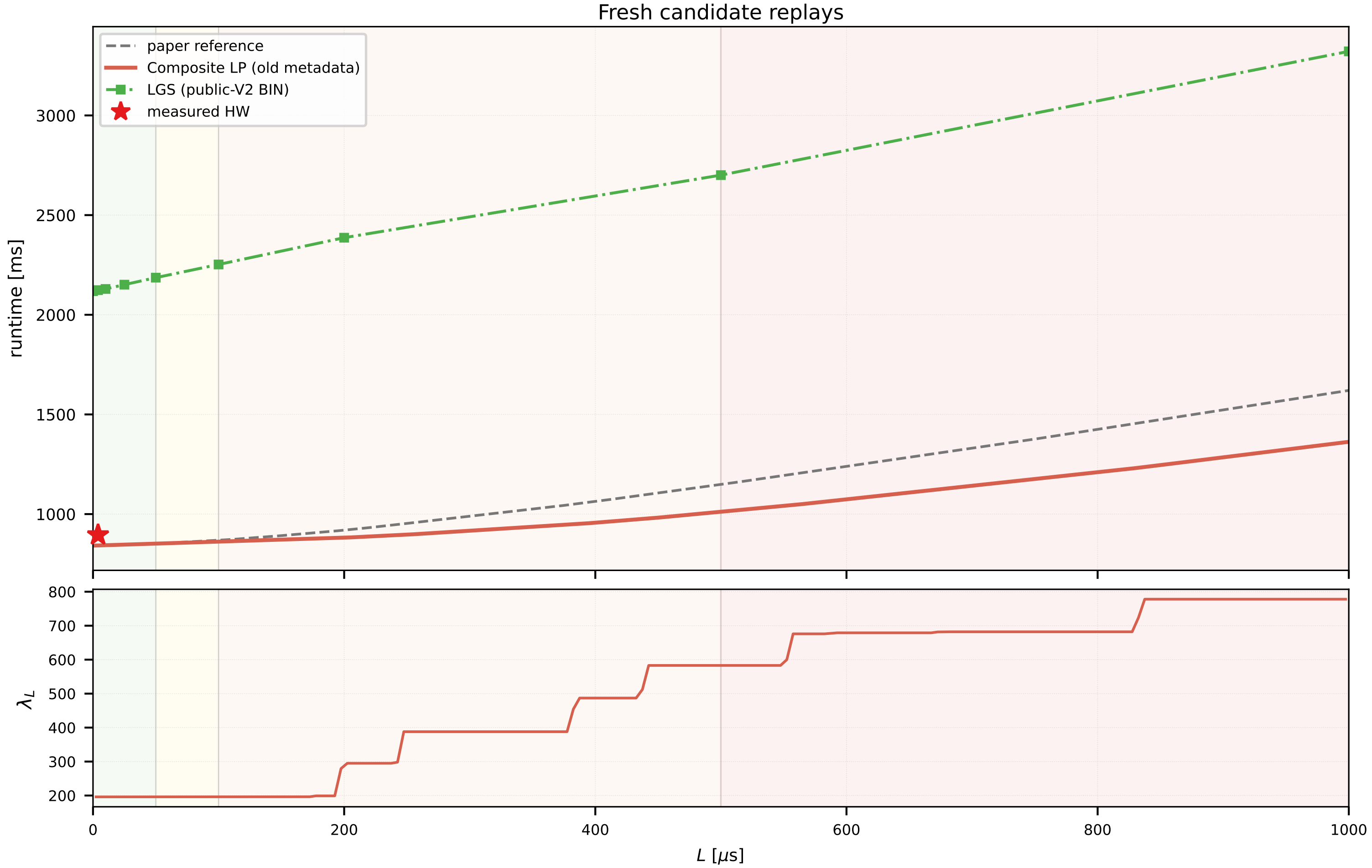
Job 1808340 is fully end-to-end from 64 recovered NSYS reports and is the numerical curve source (Monolithic max error 0.118%). Its 0.5–256 MiB messages conflict with the paper text. Corrected job 1791883 matches the written 16–64 MiB/seed 20262014 configuration but is ~3× slower and does not produce the published curve.

End-to-end path for job 1808340: 64 original NSYS → fresh SQLite → fresh GOAL/metadata → fresh LGS/Composite/Monolithic solves → this plot. Packaged CSVs are used only for the dashed reference and red measured-HW star.

Figure 5 — Llama training iteration

Paper/reference on the left · recovered candidates on the right; exact paper input is still missing



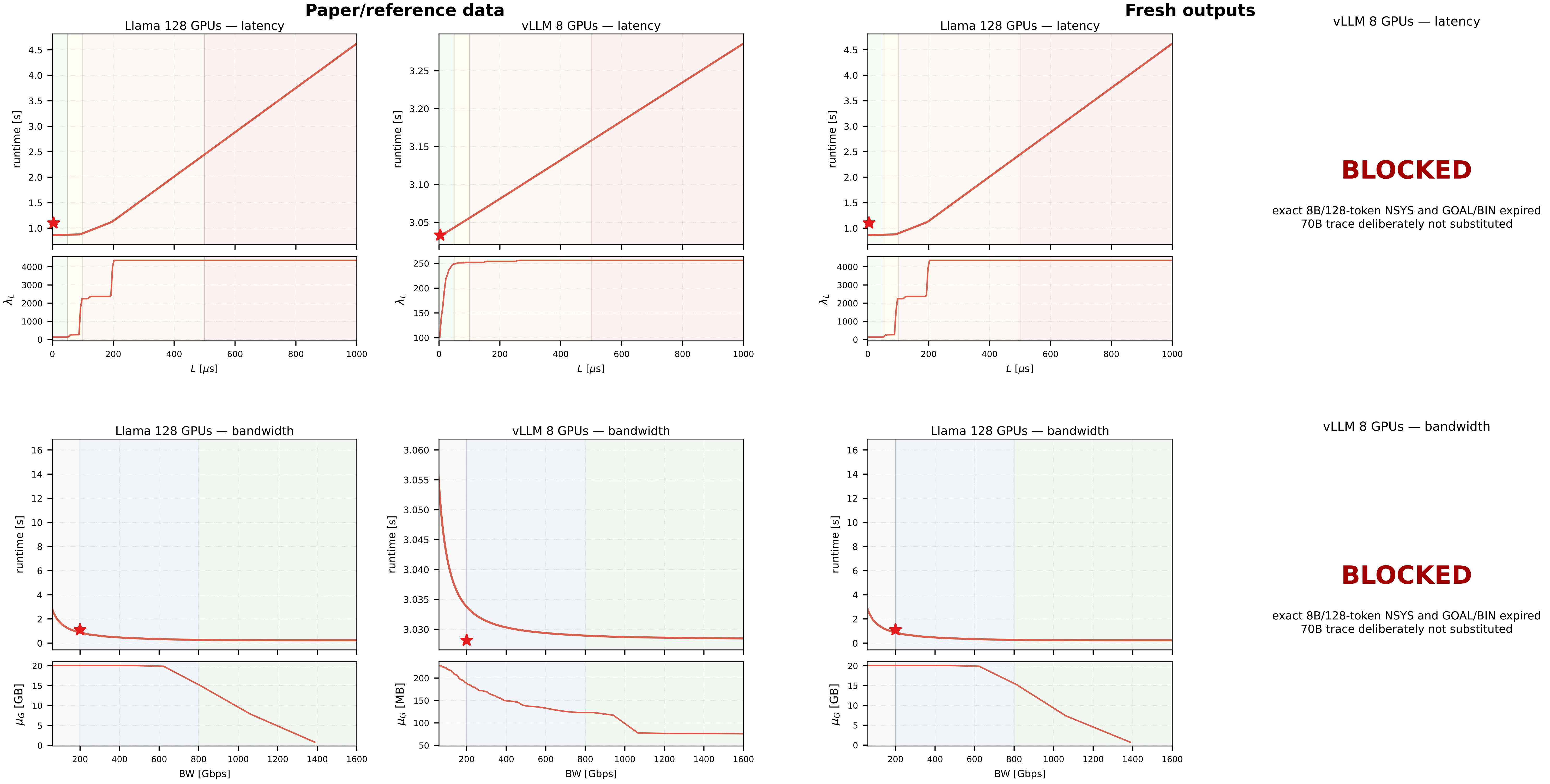


Exact paper input: unresolved
Both candidates identify Llama 7B, N4/GPU16, DP16
Paper label: Llama 8B
Raw one-iteration NSYS/GOAL: missing
Old metadata matches an older dev curve, not the paper slope
Public-V2 BIN is a different 16-rank program

The fresh old-metadata Composite starts at the same baseline as the paper but differs by 10.038% mean over the full curve. The public-V2 static BIN runs at roughly 2.1–3.3 s and is related provenance only. Neither candidate is promoted to exact.

Figure 6 — non-Grok panels

Paper/reference on the left · fresh reproduction on the right; Grok excluded as requested



Llama fresh regeneration starts from packaged NCCL metadata sidecars because no original N32 NSYS/GOAL/BIN survived locally. Every source identifies Llama 7B, while the paper labels 70B. The exact vLLM 8B job is known (1812656) but its original files expired.